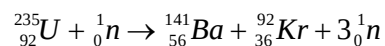


7 An induced nuclear fission reaction may be represented by the equation



(a) Sketch the variation with nucleon number of the binding energy per nucleon. [2]

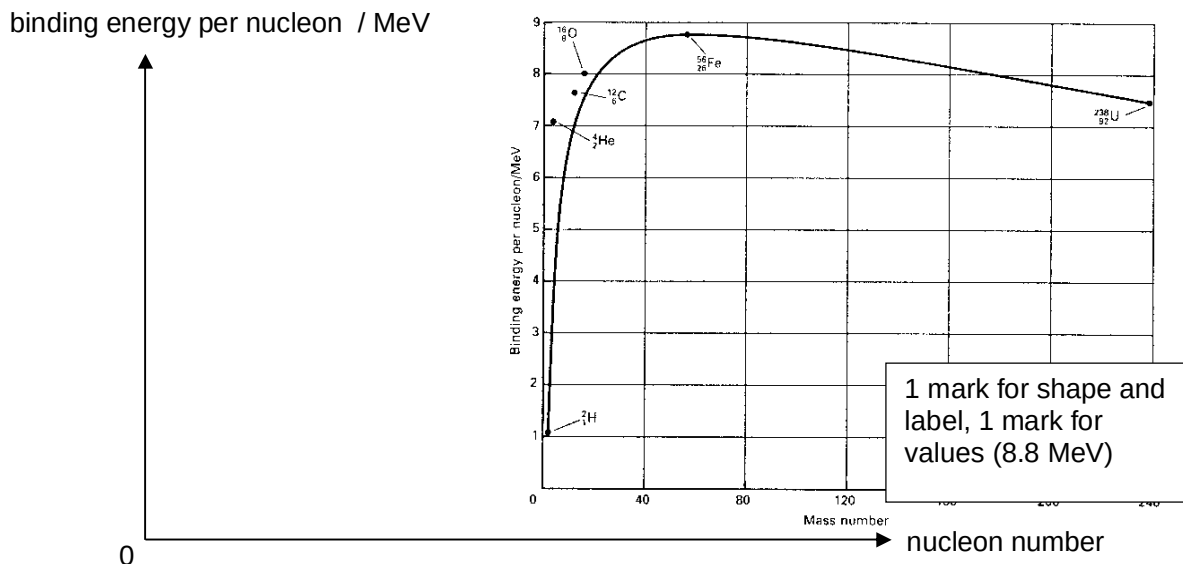


Fig. 7.1

(b) Hence, explain why Uranium – 235 is more likely to undergo fission than fusion.

Uranium-235 is on the right hand side of graph, beyond Fe-56.

BE is the BE per nucleon multiply by the number of nucleons. [B1]

Fission involves breaking a larger nucleus into smaller nuclei [B1].

Fission will involve products with higher BE per nucleon, hence more stable products and more likely to occur. [B1] or Fission implies increase of total BE and hence release of energy, hence more likely to occur. [B1]

..[3]

- (c) The masses of the various nuclides are as listed below:

nuclide	mass / u
Uranium - 235	235.04393
Barium – 141	140.91441
Krypton - 92	91.92616
neutron	1.00867

Determine the energy released in the reaction.

$$\begin{aligned}
 \text{Energy released} &= \text{mass of reactants} - \text{mass of products} \\
 &= (235.04393 + 1.00867) - [140.91441 + 91.92616 + 3(1.00867)]uc^2 \text{ [M1]} \\
 &= 0.18602uc^2 \\
 &= 2.78 \times 10^{-11} \text{ J [A1]}
 \end{aligned}$$

energy released = J [2]

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Section B

*For
Examiner's
Use*

Answer **one** question from this section in the spaces provided.

8 (a)(i) Define gravitational field strength